

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012	PSO1	PSO2	PSO3
CO1	1	2	1	2	1	1	1	2	1	1	1	2	2	2	2
CO2	2	2	2	2	2	-	2	-	2	2	2	2	3	2	3
CO3	2	2	2	2	2	-	2	-	2	2	2	2	2	2	3
CO4	2	2	3	3	2	-	2	-	2	2	2	2	2	2	3
AVG.	1.75	2.00	2.00	2.25	1.75	0.25	1.75	0.50	1.75	1.75	1.75	2.00	2.25	2.00	2.75

Machine Learning Lab

Code: PCC-CS593

Contacts: 4P

Practical: 4 hrs. / Week

Credit points: 2

Course Objective:

1.	To understand the basic theory underlying machine learning.
2.	To be able to formulate machine learning problems corresponding to different applications.
3.	To understand a range of machine learning algorithms along with their strengths and weaknesses.
4.	To be able to apply machine learning algorithms to solve problems of moderate complexity.
5.	To apply the algorithms to a real-world problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models.

Pre-Requisite:

1.	Statistics, Probability, Linear algebra, calculus and
2.	Programming language (Python)

Laboratory Experiments:

1.	Study some python libraries associated with ML
2.	Write a program in python to implement linear regression in real life dataset
3.	Write a program in python to implement logistic regression in real life dataset
4.	Write a program in python to implement k-nearest neighbour algorithm to classify a real-life dataset.
5.	Write a program in python to implement Naive based algorithm in real life dataset classification.
6.	Write a program in python to demonstrate the working of decision-tree algorithm. Use an appropriate dataset for building the decision trees and apply this knowledge to classify a new sample.
5.	Write a program to demonstrate the working of SVM algorithm. Use an appropriate dataset and apply this knowledge to classify the new sample.
6.	Write a program to implement random-forest algorithm to classify real life dataset.
7.	Write a program to implement k-means algorithm to classify real life dataset.
8.	Write a program to implement principal component analysis (PCA) in real life dataset.

9.	Write a program to implement random-forest algorithm to classify real life dataset.
10.	Write a program to implement ANN algorithm to classify real life dataset.

(Detailed instructions for Laboratory Manual to be followed for further guidance)

Text book and Reference books:

1. Manaranjan Pradhan and U Dinesh Kumar, machine Learning using Python, First edition, Wiley India Pvt. Ltd, 2019.
2. Andreas C.Mueller, Sarah Guido, Introduction to Machine Learning with Python guide for data scientists, O'REILLY Media, 2016.
3. Abhishek Vijayvargia, Machine Learning with Python, BPB Publications, 2018
4. Practical handbook of ML by S. Bhattacharya & S. Bhattacharya – G. K. Publication

Course Outcomes:	
On completion of the course students will be able to	
PCC-CS 593.1	Understand different machine learning techniques.
PCC-CS 593.2	Apply the knowledge of machine learning in various practical fields.
PCC-CS 593.3	Analyse the performances of different machine learning algorithms.
PCC-CS 593.4	Develop machine learning models to solve real life problems.

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	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012	PS01	PS02	PS03
C01	3	3	3	3	-	-	-	-	-	1	-	1	3	3	3
C02	3	3	3	3	-	-	-	-	1	-	-	2	3	3	3
C03	3	3	3	3	2	-	-	-	1	-	-	2	3	3	3
C04	3	3	3	3	3	-	-	-	1	1	-	2	3	3	3
AVG.	3.00	3.00	3.00	3.00	2.75	0.00	0.00	0.00	1.00	1.00	0.00	1.67	3.00	3.00	3.00